

ASSIGNMENT 1

Bitcoin Basics

1. Bitcoin was introduced by whom?

Answer: Satoshi Nakamoto

2. Bitcoin transactions are recorded on which data structure?

Answer: Blockchain

3. Bitcoin uses which consensus mechanism?

Answer: Proof-of-Work

4. The smallest unit of Bitcoin is called?

Answer: Satoshi (1 BTC = 100,000,000 satoshis)

5. Bitcoin is a peer-to-peer type of what system?

Answer: Network System

Wallets

6. A wallet stores the user's _____ key.

Answer: Private

7. The recovery phrase used in wallets is called a _____ phrase.

Answer: Seed

8. Hardware wallets store keys _____ (online/offline).

Answer: Offline

9. Bitcoin addresses are derived from _____ keys.

Answer: Public

10. What is required to sign a transaction? (public/private)

Answer: Private

Security

11. Bitcoin prevents double-spending using _____.

Answer: Blockchain

12. Bitcoin is secured using _____ cryptography (asymmetric/symmetric).

Answer: Asymmetric

13. To run a Bitcoin node, the software required is Bitcoin _____.

Answer: Core

14. The process of verifying transactions is called _____.

Answer: Mining

15. The Bitcoin network is described as _____ (centralized/decentralized).

Answer: Decentralized

Mining

16. Miners solve _____ puzzles to add blocks.

Answer: Cryptographic

17. Mining rewards a block with newly created _____.

Answer: Bitcoin

18. The block time target for Bitcoin is _____ minutes.

Answer: 10

19. Mining difficulty adjusts every _____ blocks.

Answer: 2016

20. Miners compete to find a valid block _____ (hash/signature).

Answer: Hash

Blockchain Structure

21. Each block contains a _____ root.

Answer: Merkle

22. Blocks are linked using the previous block's _____.

Answer: Hash

23. A Bitcoin transaction contains inputs and _____.

Answer: Hash

24. Unspent transaction outputs are called _____.

Answer: UTXOs

25. The first-ever Bitcoin block is called the _____ block.

Answer: Genesis

Network

26. A full node stores the entire blockchain _____.

Answer: Locally

27. Bitcoin uses which protocol for communication? (TCP/P2P)

Answer: P2P

28. A light wallet uses _____ mode (SPV/full).

Answer: SPV

29. Nodes relay information across a _____ network.

Answer: Peer-to-peer

30. The maximum Bitcoin supply is _____ million.

Answer: 21

B1. Explore a Block

The following information was noted from the latest Bitcoin block:

Information	Answer
Block Height	961,840
Block Hash	00000-051af
Timestamp	August 10, 2026, 05:43:34 AM UTC
Number of Transactions	4,758
Miner/Pool Name	Foundry USA

B2. Explore a Transaction

The following Bitcoin transaction was selected:

TXID:

c1742c7511a614087e6cdd876cd04018cd4de9ebdb1ffc9ec28f9b998b093787

Information	Answer
TXID	c1742c7511a614087e6cdd876cd04018cd4de9ebdb1ffc9ec28f9b998b093787
Number of Inputs	2
Number of Outputs	2
Transaction Fee	10,000 satoshis
Confirmations	Approximately 600,811

B3. Explore an Address

15. Bitcoin Address

Address:

1M37hAk6zqANwkHYyAAytm1xsczcj8pAtR

16. Address Information

Information	Answer
Total BTC Received	12.68453217
Total BTC Sent	12.35184729
Current Balance	0.33268488

What is the role of a full node?

A full node independently downloads and verifies Bitcoin blocks and transactions according to the network's consensus rules. It allows users to verify the Bitcoin network without having to trust miners or another third party.

List any two developer resources available on Bitcoin.org.

1. **Bitcoin Developer Documentation**
2. **Bitcoin Core Documentation**

Part C: Explore CoinMarketCap

C1. Market Information

The following Bitcoin market information was recorded:

Market Information	Answer
Current Price of BTC	\$65,131.17 USD
24-Hour Change	+0.58%
Market Capitalization	\$1.3 trillion USD
Circulating Supply	20.06 million BTC
Total Supply	20.06 million BTC
Maximum Supply	21 million BTC

C2. Historical Data

1. What was the price of Bitcoin one year ago?

Answer: \$119,306.76 USD

2. What were the highest and lowest prices in the last 30 days?

Highest Price: Approximately \$65,401.69 USD

Lowest Price: Approximately \$60,000 USD

Part D: Explore Investopedia

Why is Bitcoin considered volatile?

Bitcoin is considered volatile because its price can change significantly in a short period. Factors such as market sentiment, speculation, social-media influence, large transactions by major holders, limited liquidity, regulatory news, and 24-hour trading can cause large price movements.

List two investment risks mentioned.

1. **Price Volatility**
2. **Regulatory Uncertainty**

Part E: Reflection & Analysis

Which website gave you the clearest understanding? Why?

Answer: Bitcoin.org gave me the clearest understanding because it explains Bitcoin's basic concepts in simple and organized language. It helped me understand important topics such as wallets, private keys, transactions, mining, blockchain, nodes, and the peer-to-peer network.

How does blockchain ensure trust without a central authority?

Blockchain ensures trust without a central authority by maintaining a shared record of transactions across a decentralized network of computers called nodes. Each node verifies transactions according to predefined rules, while cryptographic hashes connect blocks together. Bitcoin's proof-of-work mechanism makes it computationally difficult to alter previous transactions or create invalid blocks. Because many independent participants verify the same information, users do not need to depend on a bank or other central authority to maintain the transaction record. This combination of decentralization, cryptography, proof-of-work, and independent verification allows the Bitcoin network to maintain trust without a central authority.